

Continuous Computation

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CHAPTER I
Measure Theory

1.1 Measure

Measure and integration theory develops the necessary tool for solving PDEs and stochastic processes.

σ -Algebra

Motivation In physics, the task of finding “volume” is ubiquitous. For instance in computing mass, charge, the integral would resemble:

$$\int (\cdot) dV \quad (1)$$

In some sense, this dV is a function $dV : E \rightarrow \mathbb{R}$ which computes the volume of $E \subset \mathbb{R}^n$, a tiny control volume we are interested in. It would be very natural to ask for this function to satisfy:

Definition 1.1 (Measure)

Suppose X is a set. Let $\mu : A \rightarrow \mathbb{R}$ where $A \in P(X)$ such that:

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$$\mu(\emptyset) = 0 \quad (2)$$

- **(countable additivity)** $E_1, E_2, \dots$ is a countable infinite sequence of disjoint, then:

$$\mu(\cup_j E_j) = \sum_j \mu(E_j) \quad (3)$$

Then we call μ a measure.

The question would be, when can we find such a function. Trivially we can obtain such property by sending every A to 0. However, for more practical purposes of modeling the world, we would like nicer properties. In particular, let's consider this completely reasonable demand on $\mathbb{R}^n$.

Exercise 1.1 (Open Set as a Topology)

1. Let $\mathcal{B}$ be the family of all open intervals in $\mathbb{R}$, i.e. subsets of the form

$$(a, b) := \{c \in \mathbb{R} \mid a < c < b\} \quad (4)$$

for all $a, b \in \mathbb{R}$ with $a < b$. Let $\mathcal{T}(\mathcal{B})$ be the family of all arbitrary unions of elements in $\mathcal{B}$, i.e.,

$$\mathcal{T}(\mathcal{B}) := \left\{ \bigcup_{B \in \mathcal{B}'} B \mid \mathcal{B}' \subseteq \mathcal{B} \right\} \quad (5)$$

Prove that $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$. (This “ $\mathcal{B}$ ” stands for a “base” = “basis” of a topology.)

2. For $\mathcal{B}$ and $\mathcal{T}(\mathcal{B})$ as above and any $U \in \mathcal{T}(\mathcal{B})$ (i.e., any open set), prove that there exists a countable family $\mathcal{B}'' \subseteq \mathcal{B}$ (of open intervals) such that $U = \bigcup_{B \in \mathcal{B}''} B$. (Hint: it might help to prove first that the union of any family of open intervals containing a given $x \in \mathbb{R}$ is a generalized open interval in the sense that it is of the form (a, b) , $(-\infty, a)$ or (a, ∞) for some $a, b \in \mathbb{R}$. Or do this any other way.)
3. For $\mathcal{T}(\mathcal{B})$ as above, prove that $\mathcal{T}(\mathcal{B}) \subseteq \mathcal{B}_{\mathbb{R}}$, the Borel σ -algebra on $\mathbb{R}$. (This “ $\mathcal{B}$ ”, confusingly, stands for “Borel”.)
4. Use the above to give a full proof of Proposition 1.2 on p. 22 in [F].
5. Also deduce that $\mathcal{M}(\mathcal{B}) = \mathcal{M}(\mathcal{T}(\mathcal{B})) = \mathcal{B}_{\mathbb{R}}$.

Solution to Exercise 1.1

1. We prove that $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$.

- **Property 1:** $\emptyset, \mathbb{R} \in \mathcal{T}(\mathcal{B})$

- $\emptyset = \cup_{B \in \emptyset} B$ (empty union), so $\emptyset \in \mathcal{T}(\mathcal{B})$
- $\mathbb{R} = (-\infty, \infty)$ is a generalized open interval, hence $\mathbb{R} \in \mathcal{T}(\mathcal{B})$. More explicitly, we can cover it by $\cup_{a=0}^{\infty} \{(a, a+1), (-a-1, -a)\}$

- **Property 2: Closed under arbitrary unions**

- Let $\mathcal{U} \subseteq \mathcal{T}(\mathcal{B})$ be an arbitrary collection of sets in $\mathcal{T}(\mathcal{B})$. For each $U \in \mathcal{U}$, we have $U = \cup_{B \in \mathcal{B}_U} B$ for some $\mathcal{B}_U \subseteq \mathcal{B}$. Then $\cup_{U \in \mathcal{U}} U = \cup_{U \in \mathcal{U}} (\cup_{B \in \mathcal{B}_U} B) = \cup_{B \in \cup_{U \in \mathcal{U}} \mathcal{B}_U} B$. Since $\cup_{U \in \mathcal{U}} \mathcal{B}_U \subseteq \mathcal{B}$, we have $\cup_{U \in \mathcal{U}} U \in \mathcal{T}(\mathcal{B})$

- **Property 3: Closed under finite intersections**

- By induction on the number n of sets.
- **Base case** ($n = 1$): Trivial, $U \in \mathcal{T}(\mathcal{B})$.
- **Inductive step:** Assume $U_1 \cap \cdots \cap U_n \in \mathcal{T}(\mathcal{B})$ for any n sets in $\mathcal{T}(\mathcal{B})$. For $n + 1$ sets, write $(U_1 \cap \cdots \cap U_n) \cap U_{n+1}$. By induction hypothesis, $U_1 \cap \cdots \cap U_n = \cup_{i \in I} (a_i, b_i)$ for some countable collection of open intervals. Then

$$(U_1 \cap \cdots \cap U_n) \cap U_{n+1} = (\cup_{i \in I} (a_i, b_i)) \cap U_{n+1} = \cup_{i \in I} ((a_i, b_i) \cap U_{n+1}) \quad (6)$$

Each $(a_i, b_i) \cap U_{n+1}$ is an intersection of an open interval with a union of open intervals, which is a finite union of open intervals (or empty). Hence $(U_1 \cap \cdots \cap U_{n+1}) \in \mathcal{T}(\mathcal{B})$.

Therefore $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$.

2. Suppose $\mathcal{B}' \subseteq \mathcal{B}$ and

$$U = \bigcup_{B \in \mathcal{B}'} B \quad (7)$$

where $\mathcal{B}'$ can be uncountable. We would like to show $\exists \mathcal{B}''$ that is countable such that

$$U = \bigcup_{B \in \mathcal{B}''} B \quad (8)$$

We now show a specific way to construct $\mathcal{B}''$ from $\mathcal{B}'$. Define:

$$\mathcal{B}_x = \{(a, b) \in \mathcal{B}' : x \in (a, b)\} \quad (9)$$

Clearly $\mathcal{B}_x \subseteq \mathcal{B}'$. Now proceed with the following procedure (an “infinite algorithm”) on the construction of $\mathcal{B}''$

Algorithm 1.1**Input:** A potentially uncountable set of open intervals $\mathcal{B}'$ such that $\cup \mathcal{B}' = U$ **Output:** A countable set of open intervals $\mathcal{B}''$ such that $\cup \mathcal{B}'' = U$

1. $\mathcal{B}'' = \emptyset$
2. **While** $\exists x \in U, x \notin \cup \mathcal{B}''$:
 - (a) Compute $(c, d) = \bigcup_{B \in \mathcal{B}_x} B$ where c, d is in extended real number system.
 - (b) **If** $c, d \in \mathbb{R}$:
 - i. $\mathcal{B}'_x = \{(c, d)\}$
 - (c) **Else if** $c \in \mathbb{R}, d = \infty$:
 - i. $\mathcal{B}'_x \leftarrow \{(c, [c])\} \cup \{[c], [c] + j\}_{j=1}^{\infty}$
 - (d) **Else** Other cases follows as (3)
 - (e) $\mathcal{B}'' = \mathcal{B}'' \cup \mathcal{B}'_x$
3. **Return** $\mathcal{B}''$

where $[c]$ picks the closes integer towards the right of the real axis. Note we implicitly invoked the claim below in computing (c, d) :

- **Cliam 1** The union of arbitrary family of open intervals containing $x \in \mathbb{R}$ is a *generalized open interval*

$$\bigcup_{B \in \mathcal{B}_x} B \in \{(c, d) : c, d \in \mathbb{R} \cup \{\infty, -\infty\}\} \quad (10)$$

- I claim the union is just (c, d) where $c = \inf\{a : (a, b) \in \mathcal{B}_x\}$ and $d = \sup\{b : (a, b) \in \mathcal{B}_x\}$ where we allow ∞ and $-\infty$. Clearly (c, d) covers $\cup \mathcal{B}_x$. It suffice to show $\cup \mathcal{B}_x$ covers (c, d) . Suppose $\exists y \in (c, d)$ and $y \notin \cup \mathcal{B}_x$. This means $\cup \mathcal{B}_x$ has a hole and there must exists a pair of open interval in $\mathcal{B}_x$ that is disjoint, contradicting that all open intervals in $\mathcal{B}_x$ contains x (in that case no pair should be disjoint).

Now what's left to show is (1) Only countable number of $\mathcal{B}'_x$ is needed. (2) each $\mathcal{B}'_x$ is countable. The later is obvious by construction. The former can be argued by countability of $\mathbb{Q}$. Pick a $q \in \mathbb{Q}$ in one of the open interval of $\mathcal{B}'_x$, this is always possible by density of rationals. Suppose uncountable $\mathcal{B}'_x$ is needed to cover $\cup \mathcal{B}''$ then $U = \cup \mathcal{B}''$ would contain uncountable number of rationals. This finishes the proof.

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3. By definition, $\mathcal{B}_{\mathbb{R}}$ is the smallest σ -algebra containing $\mathcal{T}(\mathcal{B})$. Therefore the inclusion holds by construction.
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4. We only need to show (b), (c), and (e) since (a) and (d) are generalized open intervals and are actually shown by (5).

Case (b): Closed intervals $\mathcal{E}_2 = \{[a, b] : a < b\}$

- $\mathcal{M}(\mathcal{E}_2) \subseteq \mathcal{B}_{\mathbb{R}}$: Note that $[a, b] = \cap_{j=1}^{\infty} (a - 1/j, b + 1/j)$ is a countable intersection of open intervals, so $\mathcal{E}_2 \subset \mathcal{B}_{\mathbb{R}}$.
- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_2)$: $(a, b) = \cup_{j=1}^{\infty} [a + 1/j, b - 1/j]$ is a countable union of closed intervals, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_2)$ and hence $\mathcal{B}_{\mathbb{R}} = \mathcal{M}(\mathcal{B}) \subset \mathcal{M}(\mathcal{E}_2)$.

Case (c): Half-open intervals $\mathcal{E}_3 = \{(a, b] : a < b\}$

- $\mathcal{M}(\mathcal{E}_3) \subseteq \mathcal{B}_{\mathbb{R}}$: $(a, b] = (a, b) \cup \{b\}$ where $(a, b) \in \mathcal{B}_{\mathbb{R}}$ and $\{b\} = \cap_{j=1}^{\infty} (b - 1/j, b + 1/j) \in \mathcal{B}_{\mathbb{R}}$.
- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_3)$: $(a, b) = \cup_{j=1}^{\infty} (a, b - 1/j]$ is a countable union of half-open intervals, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_3)$ and hence $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_3)$.

Case (e): Closed rays $\mathcal{E}_7 = \{[a, \infty) : a \in \mathbb{R}\}$

- $\mathcal{M}(\mathcal{E}_7) \subseteq \mathcal{B}_{\mathbb{R}}$: $[a, \infty) = \cap_{j=1}^{\infty} (a - 1/j, \infty)$ is a countable intersection of open rays (generalized open intervals), so $\mathcal{E}_7 \subset \mathcal{B}_{\mathbb{R}}$.
- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_7)$: $(a, b) = (a, \infty) \cap (-\infty, b)$ where $(a, \infty) = \cup_{j=1}^{\infty} [a + 1/j, \infty)$ and $(-\infty, b) = \mathbb{R} \setminus [b, \infty)$ are in $\mathcal{M}(\mathcal{E}_7)$, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_7)$ and hence $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_7)$.

Cases (c'), (e') follow by symmetry.

5. We only need to show

$$\begin{aligned} \mathcal{B} &\subseteq \mathcal{M}(\mathcal{T}(\mathcal{B})) \\ \mathcal{T}(\mathcal{B}) &\subseteq \mathcal{M}(\mathcal{B}) \end{aligned} \tag{11}$$

- The former is trivial since $\mathcal{B} \subseteq \mathcal{T}(\mathcal{B}) \subseteq \mathcal{M}(\mathcal{T}(\mathcal{B}))$
- The later can be shown using (2). $\mathcal{M}(\mathcal{B})$ contains all countable unions of open intervals $\mathcal{B}$ while $\mathcal{T}(\mathcal{B})$ contains arbitrary union. (2) bridges the gap by claiming the arbitrary union can actually be replaced by countable union.

Construction of Measure

The big picture The general idea of constructing measure is, in crude terms:

1. Find a collection of set that has a clear notion of size
2. Approximate arbitrary subsets using element in the elementary set, hence get a notion of size on arbitrary subsets
3. Make the notion of size truly a measure by restricting to a special collection of subset.

Observe how the notion of size ripens:

$$\begin{aligned} \text{Stage 1: } \boxed{|\cdot| : \mathcal{E} \rightarrow \mathbb{R}^+} &\longrightarrow \text{Stage 2: } \boxed{\mu^* : P(X) \rightarrow \mathbb{R}^+} \\ &\longrightarrow \text{Stage 3: } \boxed{\mu : \mathcal{M}_{\mu^*} \rightarrow \mathbb{R}^+} \end{aligned} \tag{12}$$

1. *Stage 1 to 2*: Create a notion of size on arbitrary sets $A \subset P(X)$ by approximating them using “building blocks” ($A \in \mathcal{E}$) which has a notion of size $|\cdot|$. This sacrifices additivity and we instead of countable sub-additivity
2. *Stage 2 to 3*: Shrink the domain by restrict to a particular family $\mathcal{M}_{\mu^*} \subset P(X)$. This gives us *countable additivity* and μ is now a **complete measure**

Example 1.1

Concretely, think $\mathcal{E}$ as the set of all open rectangles in $\mathbb{R}^2$:

$$\mathcal{E} = \{(a, b) \times (c, d) | a < b, c < d; a, b, c, d \in \mathbb{R}\} \quad (13)$$

These clearly “canonical” notion of size, namingly:

$$|A| = (d - c)(b - a) \quad (14)$$

Now using these as “building blocks”, we can measure arbitrary sets $E \in P(\mathbb{R}^2)$, for instance, a unit circle. Here is a quick numerical demo for approximating a unit circle denoted by E with squares on a grid. The resolution n is defined as the number of intervals in $(0, 1)$.

As we take finer and finer squares, we approach a limit(or the liminf). This number can be used as a notion of size for the set E . This motivates the definition of *outer measure*.

How did we compute that?

A simple way is to count grid points that is inside the unit circle. A quick `julia` implementation gives the above plot:

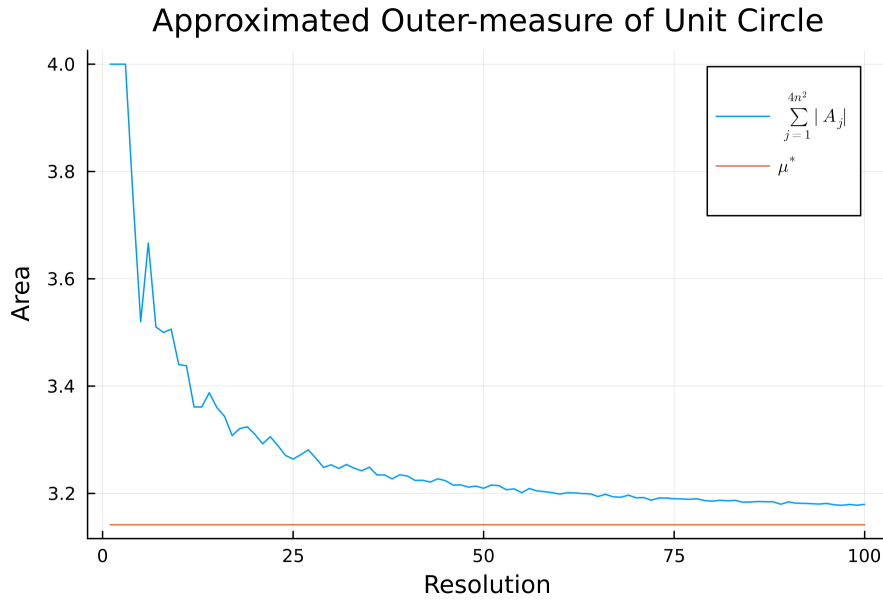
```
function compute_area_of_circle(n)
    cover::Int = 0
    for i in -n:n
        for j in -n:n
            if ((i/n)^2 + (j/n)^2) < 1
                if i == 0 && j == 0
                    cover += 4
                elseif i == 0 || j == 0
                    cover += 2
                else
                    cover += 1
                end
            end
        end
    end
    return cover * (1/n)^2
end
```

Outer Measure So how do we actually construct measures no matter what set we are working on?(Though in this book we are really interested in just $\mathbb{R}^n$). A general guideline is:

1. Find some *elementary set* $\mathcal{E}$ where premeasure is defined on the disjoint union of $\mathcal{E}$.
2. Define the measure of arbitrary subset $A \in P(X)$ as the the size of “smallest” cover of A using the elementary set:

$$\inf \left\{ \sum_{\nu=1}^{\infty} \rho(E_{\nu}) : E_{\nu} \in \mathcal{E}, A \subset \bigcup_{\nu=1}^{\infty} E_{\nu} \right\} \quad (15)$$

More generally we can define an *outer measure* as follows:



Definition 1.2 (Outer Measure)

An outer measure on a non-empty set X is a function $\mu^* : P(X) \rightarrow [0, \infty]$ such that:

- $\mu^*(\emptyset) = 0$
- If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$
- $\mu^*(\cup_{\nu} A_{\nu}) \leq \sum_{\nu} \mu^*(A_{\nu})$

We note that property (2),(3) are both derived properties of measure. So we can think the definition of outer measure as a relaxation of measure: take out countable additivity and adds back two derived property. In addition, We note that (15) obeys Definition 1.1

Exercise

Show that (15) indeed defines an outer measure Definition 1.1.

Carathéodory Family $\mathcal{M}_{\mu^*}$ How do we then obtain measure from outer measure? We note that what outer measure lacks is finite additivity, so we can further ask, is there a collection of subset $\mathcal{M} \subset P(X)$ such that μ^* is actually finitely additive? The answer is actually YES! However this seems rather arbitrary:

$$\mathcal{M} = \{A \subset X : \mu^*(F) = \mu^*(F - A) + \mu^*(A \cap F), \forall F \subset X\} \quad (16)$$

We call elements of $\mathcal{M}$ **μ^* -measureable**. In plain English, this saying A can “split measure” arbitrary $F \subset X$ by measuring the intersection and the complement separately and add them together.

Theorem 1.1 (Carathéodory’s Theorem)

If μ^* is an outer measure of X then

1. the collection of **μ^* -measureable** sets:

$$\mathcal{M} = \{A \subset X : \mu^*(F) = \mu^*(A^c \cap F) + \mu^*(A \cap F), \forall F \subset X\} \quad (17)$$

is a σ -algebra.

2. $\mu^*|_{\mathcal{M}}$ is a **complete measure**